

**PROJECT 4 – CREATING
AND SHARING
GEOSPATIAL PACKAGES
USING PYTHON AND
ARCPY**

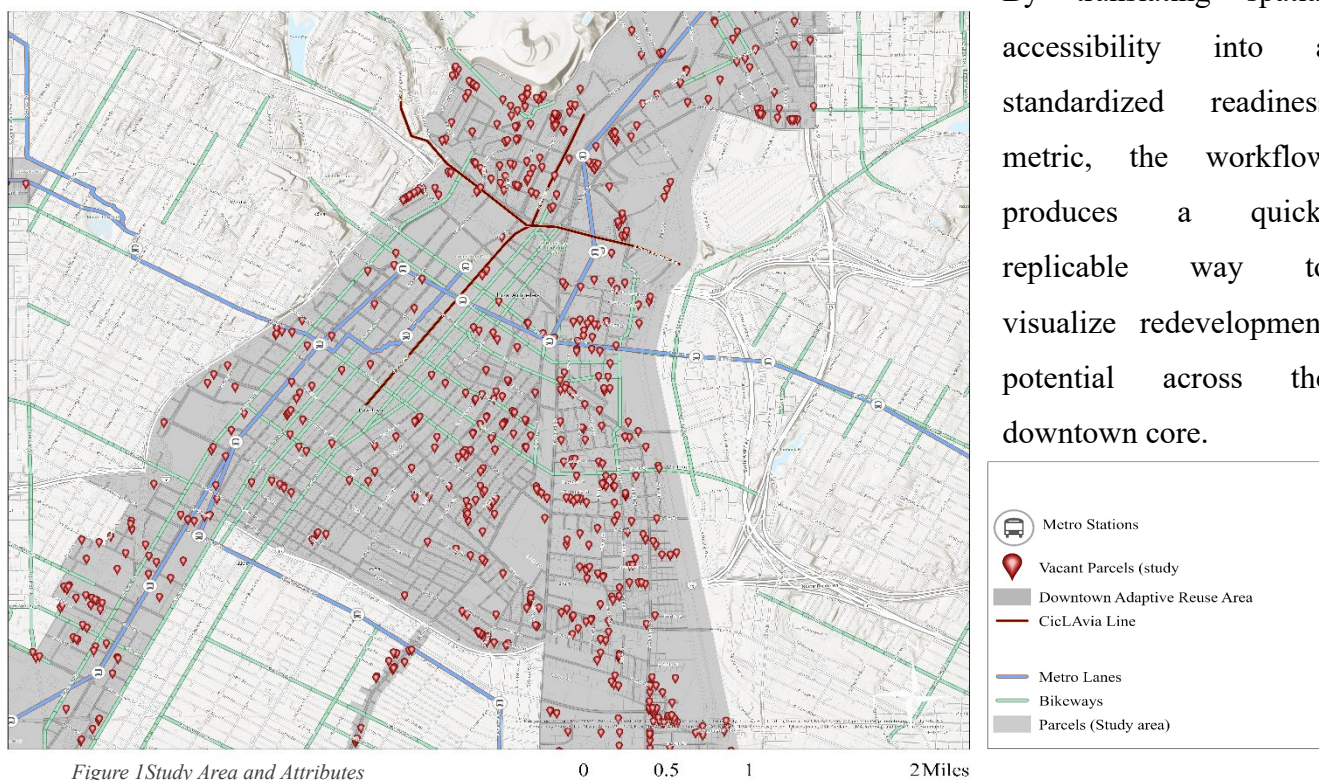
SSCI581

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The overall goal of this project is to apply advanced Python and programming skills to design, document, and share a custom geoprocessing tool within ArcGIS Pro. The assignment emphasizes creating reusable, shareable workflows that can be packaged, published, and integrated into professional GIS environments. Within this framework, my specific project focused on developing a tool that automates the evaluation of adaptive-reuse readiness for vacant parcels in Downtown Los Angeles (DTLA). Drawing from prior research and an understanding of urban form in DTLA, I designed the Vacant Reuse Buffer Scorer toolbox to identify which parcels are most spatially suitable for redevelopment based on proximity to key mobility and activity corridors. Because I had previously conducted related analyses, I already had expectations for how results should spatially look, higher readiness scores clustering around transit nodes and active transportation corridors.

This project integrates multiple open datasets: the Los Angeles City Open Data GitHub (parcels, vacant parcels, and bicycle lanes), Metro Los Angeles (rail stations and metro lines), and Open street events features using the latest CicLAvia event route in Downtown LA as an example. The tool buffers these transportation and public-activity layers by half a mile, intersects them with vacant-parcel points, and assigns a score from 0 to 3 based on the number of overlapping buffers.

By translating spatial accessibility into a standardized readiness metric, the workflow produces a quick, replicable way to visualize redevelopment potential across the downtown core.



The study area centers on Downtown Los Angeles, delimited by the Downtown Adaptive Reuse Incentive Area boundary defined in the Adaptive Reuse Ordinance (ARO) Master Plan. All datasets from GitHub and external sources were clipped to this boundary to focus the analysis and improve processing efficiency. DTLA provides an ideal testbed for adaptive-reuse modeling because it contains a high concentration of vacant parcels, a dense network of multi-modal transportation, and a long history of redevelopment projects. The ARO program has documented numerous successful conversions of abandoned commercial and industrial properties into residential or mixed-use buildings over the past two decades; this historical record serves as a benchmark for evaluating the validity of new spatial-suitability measures.

Geographically, DTLA concentrates overlapping transit corridors, including Metro A (E) Line rail segments, major bicycle lanes, and frequent CicLAvia open-street events, creating a setting where mobility accessibility and urban regeneration intersect. For these reasons, DTLA was selected as both a methodological pilot area and a conceptual case study for testing a reproducible Reuse-based scoring tool.

Data and Data Processing

The project used six primary datasets. Each was re-projected or verified in WGS 1984 / UTM Zone 11N (EPSG 32611) to maintain spatial consistency. Clipping all layers to the Downtown ARO boundary reduced processing time and ensured the results reflected only parcels relevant to adaptive-reuse policy.

Dataset	Source	Type	Coordinate System	Notes
Parcels	LA City Open Data (GitHub)	Polygon	WGS 84 / UTM 11N	Full parcel fabric for LA
Vacant Parcels	LA City Open Data (GitHub)	Point	WGS 84 / UTM 11N	Subset classified as vacant
Bicycle Lanes	LA City Open Data (GitHub)	Line	WGS 84 / UTM 11N	Bikeway network (Class I–III)

Dataset	Source	Type	Coordinate System	Notes
Metro Stations	Metro Los Angeles	Point	WGS 84 / UTM 11N	Active stations (2025 data)
Metro Rail Lines	Metro Los Angeles	Line	WGS 84 / UTM 11N	Rail corridors within DTLA
CicLAvia Route	OpenStreetMap	Line	WGS 84 / UTM 11N	Latest Downtown LA event route

Table 1. Summary of Input Data

All datasets were imported into a single project geodatabase (Reuse.gdb) to streamline processing. Pre-clipping the layers not only reduced computational load but also ensured the tool executed the results just for DTLA within ArcGIS Pro.

Methods

The Vacant Reuse Buffer Scorer toolbox was implemented as a Python Toolbox (.pyt) containing one custom tool: Buffer Hit Counter (Vacant Parcels). The tool automates an urban-planning workflow in which spatial proximity to transportation and public-space amenities is used as a proxy for redevelopment opportunity. Each input layer, bicycle lanes, Metro stations, and CicLAvia routes, is buffered by 0.5 mile. This threshold aligns with standard urban-planning metrics that consider a half-mile radius (approximately a ten-minute walk) as the maximum realistic distance of pedestrian accessibility to a node. Distances beyond half a mile are less likely to influence walk-in activity or investor interest; therefore, the buffer area effectively represents the sphere of influence of each network.

The script was created directly inside ArcGIS Pro using the built-in Python Toolbox structure. Initially, I created a blank tool from the Catalog pane and configured its default parameters (input, output, and geodatabase paths). Afterward, I edited the Python script associated with the toolbox using ArcGIS Pro's integrated code editor. This ensured parameter binding and compatibility with ArcGIS's geoprocessing environment. The workflow was then setup check the projections, buffer creation, and scoring.

Key stages of the Python implementation included:

```
class BufferHitCounter(object):
    def __init__(self):
        self.label = "Buffer Hit Counter (Vacant Parcels)"
        self.description = ("Buffers bicycle lanes, metro stations, and CicLAvia routes "
                            "by 0.5 mile (editable), then counts how many buffers each "
                            "vacant parcel POINT falls inside (0-3).")
        self.canRunInBackground = True

    def getParameterInfo(self):
        p = []

        p.append(arcpy.Parameter(
            displayName="Vacant parcels (POINTS)",
            name="vacant_pts",
            datatype="GPFeatureLayer",
            parameterType="Required",
            direction="Input"))
        p[-1].value = "VacantParcels_Clip"

        p.append(arcpy.Parameter(
            displayName="Bicycle lanes (LINES)",
            name="bike_fc",
            datatype="GPFeatureLayer",
            parameterType="Required",
            direction="Input"))
        p[-1].value = "City_of_Los_Angeles_Bikeways"

        p.append(arcpy.Parameter(
            displayName="Metro stations (POINTS)",
            name="metro_fc",
            datatype="GPFeatureLayer",
            parameterType="Required",
            direction="Input"))
        p[-1].value = "Metro Stations"

    # intro
    # 1) Copy inputs into GDB (honor selections), then project if needed
    vacant_work = _materialize(vacant_pts, "vacant"); vacant_p = _maybe_project(vacant_work, "vacant")
    bike_work = _materialize(bike_fc, "bike"); bike_p = _maybe_project(bike_work, "bike")
    metro_work = _materialize(metro_fc, "metro"); metro_p = _maybe_project(metro_work, "metro")
    ciclo_p = None
    if ciclo_fc:
        ciclo_work = _materialize(ciclo_fc, "ciclo"); ciclo_p = _maybe_project(ciclo_work, "ciclo")
```

Parameter Definitions and Validation

```
# 2) Buffers
bike_buf = _mkbuf(bike_p, "bike")
metro_buf = _mkbuf(metro_p, "metro")
ciclo_buf = _mkbuf(ciclo_p, "ciclo") if ciclo_p else None

# 4) Flags + total
for fld in ("in_bike", "in_metro", "in_ciclo", "buffer_hits"):
    _ensure_field(out_pts, fld, "SHORT")

arcpy.management.CalculateField(out_pts, "in_bike", 0, "PYTHON3")
arcpy.management.CalculateField(out_pts, "in_metro", 0, "PYTHON3")
arcpy.management.CalculateField(out_pts, "in_ciclo", 0, "PYTHON3")

arcpy.management.SelectLayerByAttribute(lyr, "CLEAR_SELECTION")
arcpy.management.SelectLayerByLocation(lyr, "INTERSECT", bike_buf, None, "NEW_SELECTION")
arcpy.management.CalculateField(lyr, "in_bike", 1, "PYTHON3")

arcpy.management.SelectLayerByAttribute(lyr, "CLEAR_SELECTION")
arcpy.management.SelectLayerByLocation(lyr, "INTERSECT", metro_buf, None, "NEW_SELECTION")
arcpy.management.CalculateField(lyr, "in_metro", 1, "PYTHON3")

if ciclo_buf:
    arcpy.management.SelectLayerByAttribute(lyr, "CLEAR_SELECTION")
    arcpy.management.SelectLayerByLocation(lyr, "INTERSECT", ciclo_buf, None, "NEW_SELECTION")
    arcpy.management.CalculateField(lyr, "in_ciclo", 1, "PYTHON3")

arcpy.management.SelectLayerByAttribute(lyr, "CLEAR_SELECTION")

arcpy.management.CalculateField(
    out_pts, "buffer_hits",
    "(!in_bike! if !in_bike! else 0) + (!in_metro! if !in_metro! else 0) + (!in_ciclo! if !in_ciclo! else 0)",
    "PYTHON3"
)
_ensure_field(out_pts, "readiness_0_100", "SHORT")
arcpy.management.CalculateField(out_pts, "readiness_0_100",
    "int(round((!buffer_hits! / 3.0) * 100))", "PYTHON3")
```

Buffer Creation and Intersection Logic

```

# 5) Summary
summary = os.path.join(out_gdb, f"{basename}_summary")
if arcpy.Exists(summary): arcpy.management.Delete(summary)
arcpy.analysis.Statistics(out_pts, summary, [{"buffer_hits", "COUNT"}], case_field="buffer_hits")

# 6) Add to map
try:
    aprx = arcpy.mp.ArcGISProject("CURRENT"); m = aprx.activeMap
    m.addDataFromPath(out_pts); m.addDataFromPath(bike_buf); m.addDataFromPath(metro_buf)
    if ciclo_buf: m.addDataFromPath(ciclo_buf)
except Exception:
    pass

arcpy.AddMessage(f"☑ Vacant points scored: {out_pts}")
arcpy.AddMessage(f"☑ Summary table: {summary}")

# utilities
def _ensure_gdb(gdb_path):
    """Create the File GDB if it doesn't exist."""
    if arcpy.Exists(gdb_path):
        return
    folder, name = os.path.split(gdb_path)
    if not os.path.isdir(folder):
        os.makedirs(folder, exist_ok=True)
    if not name.lower().endswith(".gdb"):
        name += ".gdb"
    arcpy.management.CreateFileGDB(folder, os.path.splitext(name)[0])

```

Readiness Score Calculation and Field Management

The script uses `arcpy.analysis.Buffer` to create 0.5-mile buffers around each transportation feature, applies `SelectLayerByLocation` to identify intersecting parcels, and calculates a `readiness_0_100` field scaling the buffer count (0–3) to a normalized 0–100 range. Finally, it outputs all results into the project geodatabase and automatically adds the layers back into the ArcGIS Pro map for visualization.

Results

Running the tool produced two main outputs: dissolved 0.5-mile buffer polygons for bicycle lanes, Metro stations, and CicLAvia routes, and a scored vacant-parcel layer containing all new fields and a calculated readiness score (table of attributes). The resulting attribute table lists each parcel with binary indicators of buffer inclusion (`in_bike`, `in_metro`, `in_ciclo`) and a final readiness score between 0 and 100. When symbolized with graduated colors, the parcels reveal a clear spatial hierarchy of adaptive-reuse potential. Parcels with the highest scores (2–3 buffer intersections) cluster along Broadway, Spring Street, and the Metro A Line corridor—precisely the zones where adaptive-reuse projects have historically concentrated under the ARO program.

DTLA VACANT PARCEL REUSE-SCORE

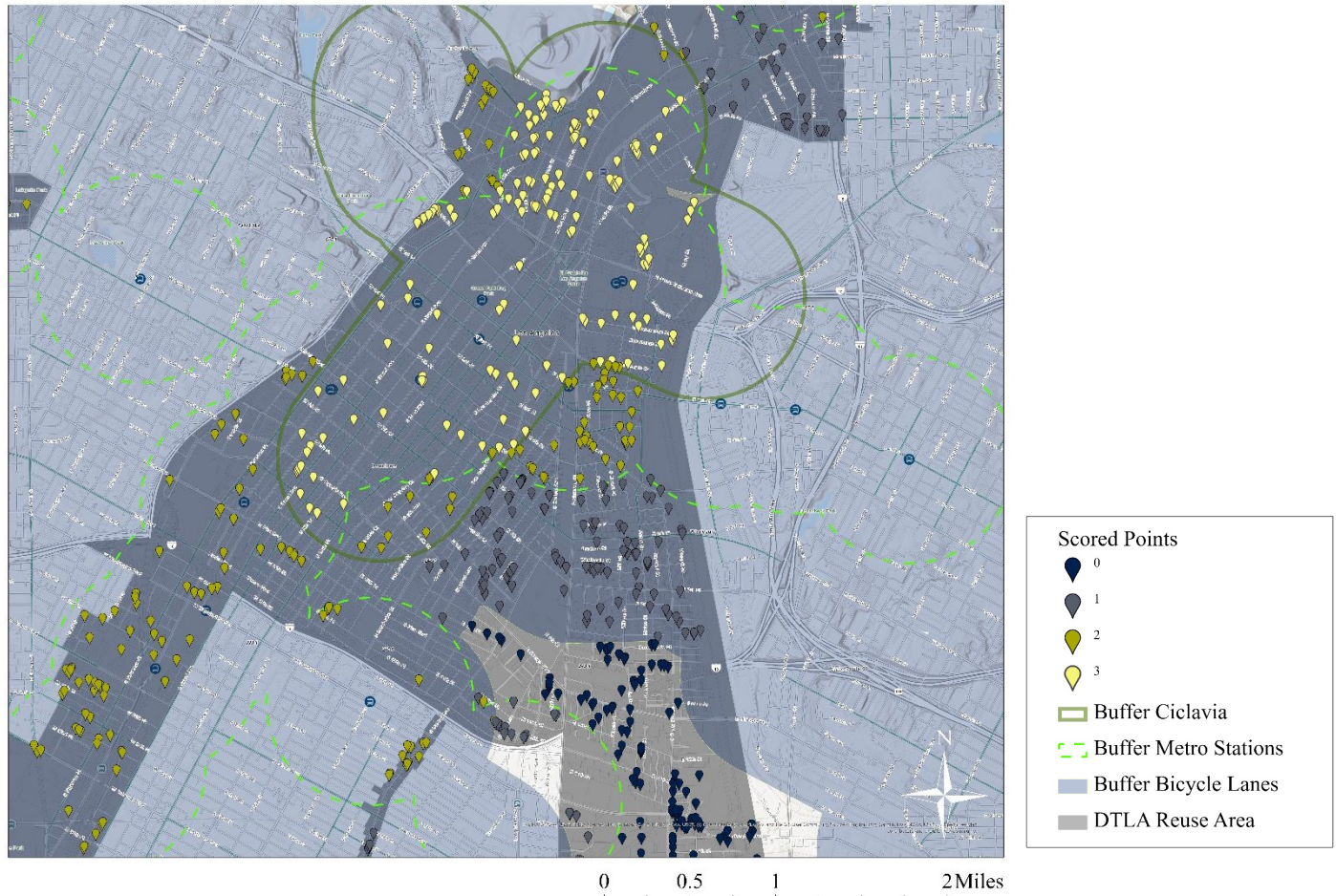


Figure 2 Results

The results confirm that high-readiness parcels align with areas already targeted for redevelopment, demonstrating the predictive logic of the tool. The buffer layers also provide visual transparency showing how each transportation or activity corridor influences parcel scoring.

One of the most valuable findings is the influence of open-street events such as CicLAvia. By integrating this layer, the analysis captures a temporal and social factor that traditional infrastructure datasets often overlook. The CicLAvia route introduces a dynamic layer of opportunity because these events temporarily convert streets into pedestrian-oriented corridors, concentrating activity in areas that may otherwise appear inactive. Vacant parcels located near these routes receive higher readiness scores, indicating that even temporary increases in walkability and public use can highlight redevelopment potential. This demonstrates how public

events like CicLAvia can stimulate new interest in underutilized areas and suggests that adaptive-reuse planning can benefit from incorporating event-based spatial data.

The results therefore not only reflect permanent infrastructural assets (bikeways and metro stations) but also episodic, human-centered factors that redefine accessibility and urban vitality. This finding expands the conceptual framework for reuse-readiness modeling by acknowledging social and cultural drivers of urban regeneration.

Discussion (Revised)

Developing this tool required addressing several technical challenges. Early in the process, parameter recognition errors and projection mismatches prevented the tool from running correctly. This was solved by implementing functions that automatically validated coordinate systems and created missing geodatabases using `arcpy.management.CreateFileGDB`. Another issue arose from ArcGIS's file locks, which prevented field creation during editing; this was fixed by reordering operations to add fields before creating feature layers. Finally, path management was difficult because the default toolbox paths conflicted with ArcGIS Pro's relative project environment, this was resolved by hardcoding and validating the workspace path using `env.workspace`. These challenges ultimately led to a more robust and reusable script that could handle common geoprocessing errors automatically. The result was a reliable, replicable tool suitable for classroom or other applications.

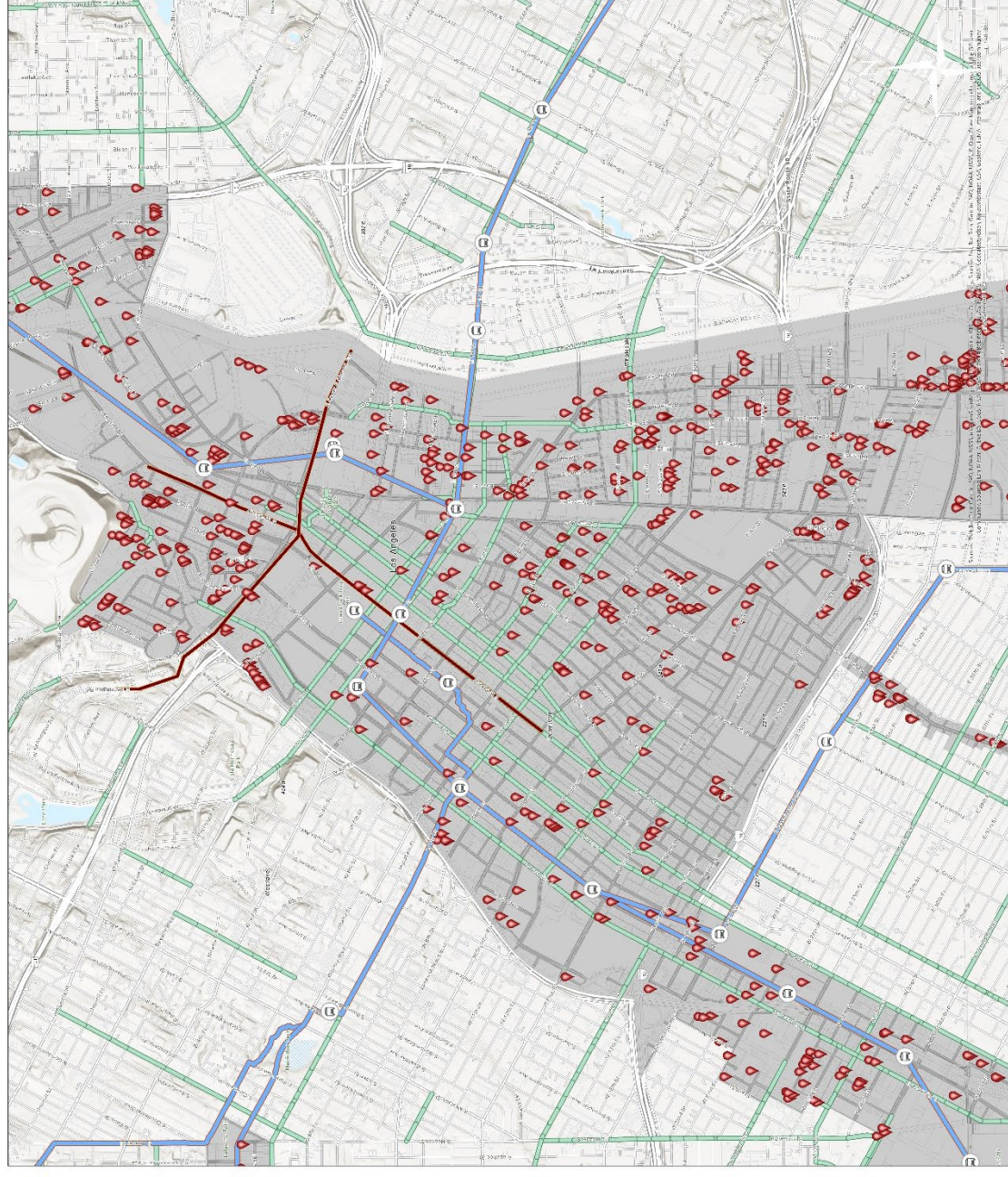
Beyond its technical success, the Vacant Reuse Buffer Scorer offers valuable potential for broader applications. The integration of CicLAvia data proved that periodic, event-based activities can meaningfully reshape accessibility metrics. Open-street initiatives temporarily close streets to cars and invite thousands of pedestrians and cyclists, transforming entire corridors into active public spaces. When analyzed spatially, these events reveal overlooked opportunities for redevelopment, especially near underused parcels and inactive commercial corridors. Incorporating similar open-street events from other cities (such as "Sunday Streets" in San Francisco, "Open Boulevards" in New York, or "Bogotá's Ciclovía") could expand the model to capture how civic events influence economic and social vitality at the neighborhood scale.

Future iterations of the tool could include additional datasets like zoning classifications, building age, or business-activity indicators to create a more comprehensive reuse index. Likewise, using network-based analysis rather than Euclidean buffers would better approximate pedestrian accessibility. Weighting variables differently, for example, assigning greater importance to Metro proximity or frequency of open-street events, would help tailor the index for different urban goals. With these enhancements, the tool could evolve into a scalable framework for measuring redevelopment readiness and supporting equitable urban revitalization.

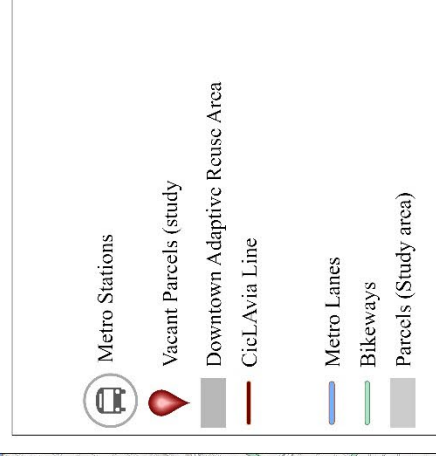
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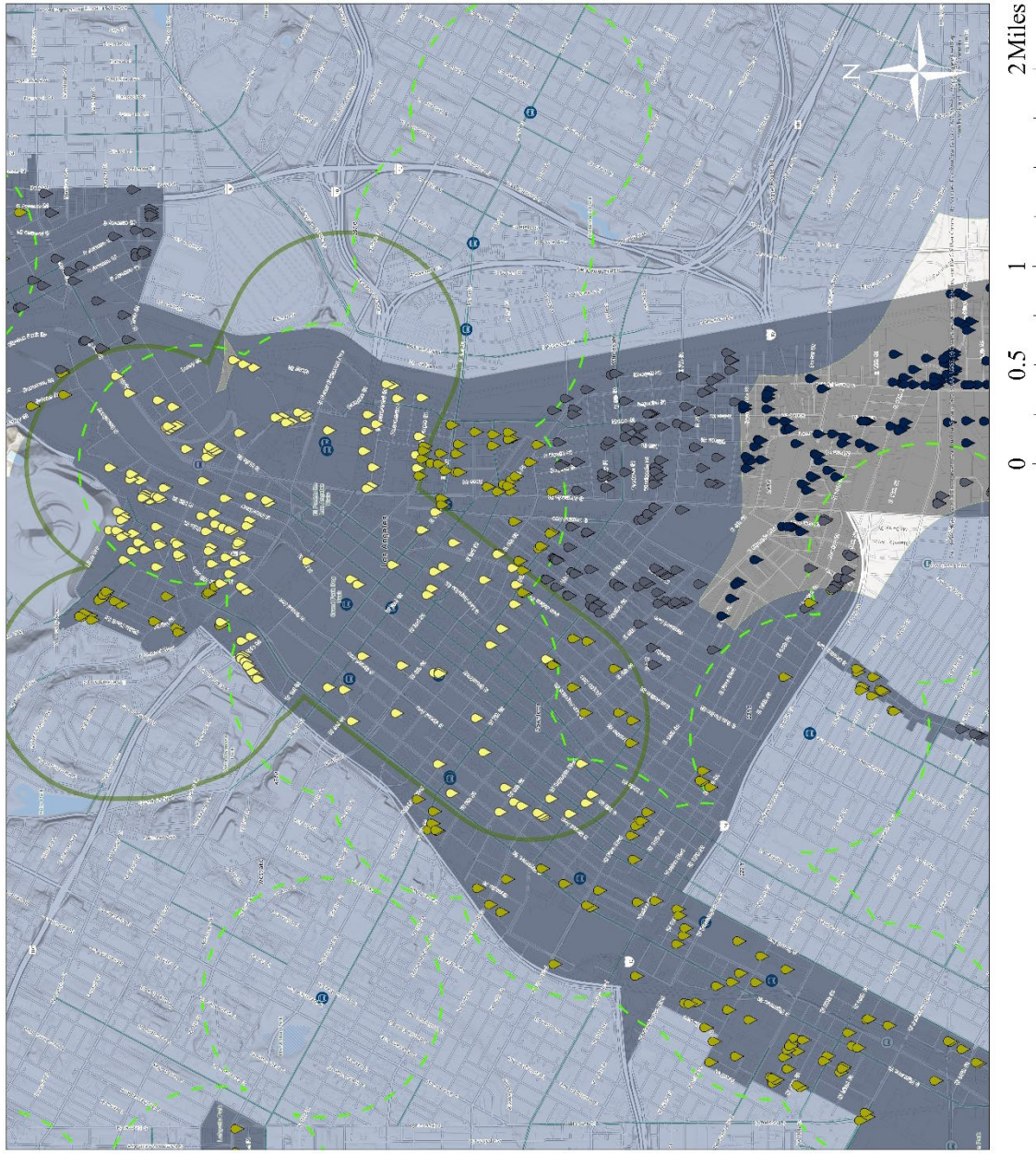
DTLA VACANT PARCEL STUDY AREA & ATTRIBUTES



This study focuses on the Downtown Los Angeles Adaptive Reuse Incentive Area, a district characterized by dense transit infrastructure, mixed land uses, and numerous underutilized parcels. The analysis integrates publicly available spatial data to identify vacant parcels with potential for redevelopment. Key layers include Metro rail stations, bicycle lanes, and CicLAvia routes, which represent major elements of sustainable urban mobility. By examining their spatial relationships, the project establishes a foundation for assessing how transportation accessibility can guide adaptive reuse strategies within the downtown core.



DTLA VACANT PARCEL REUSE-SCORE



This map identifies vacant parcels in Downtown Los Angeles that are most suitable for adaptive reuse based on proximity to multimodal transportation and open-street activity. Parcels were evaluated using a custom tool that buffers bicycle lanes, metro stations, and CicLAvia routes by 0.5 miles and counts how many of these zones intersect each parcel. The resulting reuse-score ranges from 0 (low accessibility) to 3 (high accessibility), highlighting areas with the strongest potential for redevelopment, sustainable mobility, and public-realm integration within the Downtown Reuse Incentive Area.